MOSAIC: from source data to scientifically usable records

AGENTIC SCIENTIFIC WORKFLOWS

Regional / portfolio assessment

Cross-event sampling, fragilities, regional loss

Functional recovery

Longitudinal linkage and multi-source fusion

AIM 3 Reusable expertise: skills, queries, transformations, validity checks

Shared memory: findings, failures, corrections and reasons for specification changes

records + expertise support existing agents

AIM 2 BUILDING-IN-CONTEXT RECORDS

Atomic unit: a resolvable linked-data object for an individual structure

Structure attributes + hazard intensity + site / exposure + function / criticality

Pre-event, post-event and recovery observations linked in space and time

Typed provenance: expert-assessed, machine-inferred, modeled or validated

Confidence + validation links + explicit coverage / absence states

records conform to the specification

AIM 1 SEMANTIC INTEROPERABILITY LAYER

Maintained specification: natural hazards FAIR Implementation Profile + CDIF mappings

Identifiers, meaning, structure, units, time, geography, access and conformance

describe and link in place

SOURCE DATASETS • Original repositories, formats and access conditions retained

StEER / DesignSafe | Inventories / BIM | Imagery | Hazard, site and recovery data

Evaluation evidence

Revises Aim 1

Held-out events

Solid arrows: description, conformance and use. Dashed arrow: evaluation feedback to the maintained specification.